

Secure File Sharing System Using Hill Cipher and DES

Comprehensive Academic Project Source Document & Slide Deck Blueprint

Subject: Cryptography & Network Security (CNS)	Academic Degree: B.Tech Computer Science & Engineering
Implementation: Full-Stack Web Application (FastAPI + Next.js)	Key Algorithms: Hill Cipher (Z_256) + DES-CBC + HMAC-SHA256
Document Purpose: NotebookLM Ingestion & Presentation Generation	Mathematical Ring: Integer Ring Z_256 (Byte-Level Reversible)

1. Executive Abstract & Educational Cryptography Notice

SecureVault is a production-quality, academic cryptographic file sharing system designed to demonstrate **layered defense-in-depth cryptography**. It couples a linear algebraic transformation (the polygraphic **Hill Cipher** operating over the byte ring **Z_256**) with a non-linear symmetric block cipher (the **Data Encryption Standard (DES)** operating in **Cipher Block Chaining (CBC)** mode). The cryptographic workflow is hardened with modern **PBKDF2-HMAC-SHA256** password-based key derivation and **HMAC-SHA256** Encrypt-then-MAC authentication sealed within a binary container format (.svault).

IMPORTANT EDUCATIONAL & VIVA DEFENSE ADVISORY:

This project is developed strictly for educational and pedagogical exploration of layered Feistel and matrix ciphers. **DES is cryptographically obsolete for modern production environments** due to its small 56-bit effective key length ($2^{56} \approx 7.2 \times 10^{16}$ keys), which can be brute-forced within hours by modern GPU clusters. In this academic demonstration, DES illustrates Feistel block chaining and CBC diffusion. For enterprise production applications, **AES-256-GCM** or **ChaCha20-Poly1305** must be used.

2. Problem Statement & Motivation

Single-layer cryptographic designs often introduce single points of failure. In pedagogical settings, classical ciphers are frequently studied only as textbook toy examples with grave real-world shortcomings:

- Linearity of Classical Hill Cipher:** Classical Hill Cipher operates modulo 26 on uppercase letters A–Z. Because it is a linear transformation ($C = P \cdot K \text{ mod } 26$), it is entirely vulnerable to Known-Plaintext Attacks (KPA). An adversary who intercepts n plaintext-ciphertext vector pairs can recover key matrix K using standard Gaussian elimination.
- Binary File Incompatibility:** Real-world files (PDF documents, PNG images, ZIP archives, executables) contain arbitrary 8-bit bytes spanning 0 to 255. Classical mod 26 Hill Cipher cannot process arbitrary bytes, leading to silent file truncation and catastrophic data corruption.

- **DES Block Frequency & Keyspace Limitations:** When DES is used in Electronic Codebook (ECB) mode, identical plaintext blocks generate identical ciphertext blocks, leaking structural file outlines (the ECB Penguin effect). Furthermore, DES alone has an inadequate 56-bit keyspace.
- **SecureVault's Hybrid Resolution:** SecureVault extends the Hill Cipher to the integer ring **Z₂₅₆**, enforcing mathematical invertibility for all byte values. It chains this with DES in **CBC mode** using random IVs, and seals the container using **HMAC-SHA256** under the Encrypt-then-MAC paradigm.

3. End-to-End Cryptographic Architecture

The application implements an end-to-end multi-stage pipeline where each cryptographic phase performs an orthogonal security role: key derivation, linear diffusion, non-linear confusion, and authenticated integrity.

Stage	Cryptographic Primitive	Operational Specification	Security Contribution
Stage 1: Key Derivation	PBKDF2-HMAC-SHA256 (NIST SP 800-132)	100,000 iterations, 16-byte random salt, 64 derived bytes	Strengthens low-entropy passwords; eliminates precomputed rainbow tables.
Stage 2: Layer 1 Cipher	Hill Cipher over Ring Z ₂₅₆	2x2 matrix multiplication modulo 256; vectorized via NumPy	Algebraic polygraphic diffusion across byte-pairs; destroys single-byte frequency distribution.
Stage 3: Layer 2 Cipher	DES in CBC Mode + PKCS#7	64-bit blocks, 16 Feistel rounds, 8-byte random IV	Non-linear confusion via S-boxes; CBC block chaining eliminates structural pattern leakage.
Stage 4: Authenticated Integrity	HMAC-SHA256 (Encrypt-then-MAC)	32-byte tag computed over header + ciphertext	Guarantees ciphertext integrity; detects bit-flipping; stops padding oracle exploits.

4. Mathematical Formulation: Hill Cipher in Ring Z₂₅₆

Classical Hill Cipher transforms n-tuples of alphabet letters using matrix multiplication modulo 26. SecureVault transforms pairs of arbitrary 8-bit bytes in the integer ring **Z₂₅₆**.

Encryption Vector Transformation:

Let the input binary stream be partitioned into 2-byte vectors $P = [p_1, p_2]^T$, where $p_1, p_2 \in [0, 255]$. Let K be a 2×2 key matrix with integer entries in $[0, 255]$:

$$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$
$$C = (P \cdot K^T) \bmod 256$$

Decryption Vector Transformation:

$$P = (C \cdot (K^{-1})^T) \bmod 256$$

Mathematical Proof of Invertibility in Ring $\mathbb{Z}_{256}$:

Theorem: A matrix K has a unique modular inverse $K^{-1} \bmod 256$ if and only if: $\gcd(\det(K) \bmod 256, 256) = 1$.

Proof: The prime factorization of the modulus is $256 = 2^8$. The only prime divisor of 256 is 2. Therefore, an integer is coprime to 256 if and only if it is not divisible by 2. Hence, $\gcd(\det(K), 256) = 1 \iff \det(K) \bmod 2 = 1$ ($\det(K)$ must be an ODD integer).

Guaranteed Invertible Key Derivation Construction:

SecureVault guarantees that derived Hill matrices are unconditionally invertible modulo 256 by enforcing parity: Let 4 pseudo-random bytes from PBKDF2 be mapped to entries a, b, c, d as follows:

- $a = \text{byte}[0] \mid 1$ (forced odd: $a = 2k + 1$)
- $b = \text{byte}[1] \& 0xFE$ (forced even: $b = 2m$)
- $c = \text{byte}[2] \& 0xFE$ (forced even: $c = 2n$)
- $d = \text{byte}[3] \mid 1$ (forced odd: $d = 2p + 1$)

The determinant is computed as $\det(K) = (a \cdot d - b \cdot c) \bmod 256$.

Evaluating modulo 2: $\det(K) \bmod 2 = (\text{odd} \cdot \text{odd}) - (\text{even} \cdot \text{even}) = 1 - 0 = 1$ (Always Odd!).

Because $\det(K)$ is guaranteed odd, $\gcd(\det(K), 256) = 1$ holds universally without exception.

Modular Matrix Inversion Computation:

1. Modular inverse of determinant: Solve $(\det(K) \cdot \det_{\text{inv}}) \bmod 256 = 1$ using the Extended Euclidean Algorithm.
2. Adjugate matrix: $\text{adj}(K) = \begin{bmatrix} d, -b \\ -c, a \end{bmatrix}$.
3. Modular inverse matrix: $K^{-1} = (\det_{\text{inv}} \cdot \text{adj}(K)) \bmod 256$.

5. Block Cipher Mechanics: DES in CBC Mode

The intermediate ciphertext generated by the Hill Cipher layer is passed directly into the DES engine. DES (NIST FIPS PUB 46-3) operates on 64-bit blocks using a 56-bit effective key with 16 rounds of Feistel substitution-permutation.

Cipher Block Chaining (CBC) Mechanics:

In CBC mode, each plaintext block P_i is XORed with the preceding ciphertext block C_{i-1} prior to DES encryption. The first block P_1 is XORed with an 8-byte cryptographically random Initialization Vector (IV):

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C_0 = IV (8 random bytes generated via os.urandom)
C_i = DES_Encrypt_K( P_i ⊕ C_{i-1} ), for i ≥ 1
```

CBC Decryption:

```
P_i = DES_Decrypt_K( C_i ) ⊕ C_{i-1}, with C_0 = IV
```

PKCS#7 Padding: DES operates on strict 8-byte (64-bit) blocks. Before encryption, data is padded with N bytes of value N (where $1 \leq N \leq 8$) so that total payload size is a multiple of 8. Decryption safely un pads the trailing bytes.

6. Key Management (PBKDF2) & Authenticated Integrity (HMAC)

PBKDF2 Key Derivation (NIST SP 800-132): Passwords entered by users have low entropy. SecureVault applies PBKDF2 with HMAC-SHA256, 100,000 computation rounds, and a 16-byte random salt to derive 64 pseudo-random bytes. These bytes are partitioned into:

- **Bytes 00..07 (8 Bytes):** DES 64-bit key (56 effective bits).
- **Bytes 08..39 (32 Bytes):** HMAC-SHA256 256-bit authentication key.
- **Bytes 40..43 (4 Bytes):** Hill Cipher 2×2 matrix seed coefficients.
- **Bytes 44..63 (20 Bytes):** Entropy buffer for future expansion.

HMAC-SHA256 Encrypt-then-MAC (EtM) Integrity Tag: Encryption provides confidentiality but does not guarantee authenticity. A malicious actor could flip ciphertext bits or execute padding oracle attacks. SecureVault implements Encrypt-then-MAC by calculating a 32-byte HMAC-SHA256 signature across all container metadata and ciphertext. Upon decryption, the signature is verified in **constant time** using `hmac.compare_digest` prior to invoking any DES decryption. If tampered, the operation aborts with a clean error message.

7. Binary Container Specification (.svault Format)

SecureVault encapsulates all encrypted files into a portable binary container format with the `.svault` extension:

Offset	Field Name	Length	Description
00 .. 05	Magic Header	6 Bytes	ASCII identifier b'SVAULT'
06	Format Version	1 Byte	uint8 format version (0x01)
07	Reserved Flags	1 Byte	uint8 reserved (0x00)
08 .. 23	PBKDF2 Salt	16 Bytes	Cryptographic random salt for key regeneration
24 .. 31	DES IV	8 Bytes	Random 8-byte Initialization Vector for CBC mode
32	Matrix Dim (n)	1 Byte	Matrix dimension (2 for 2×2)
33 .. 36	Hill Matrix K	4 Bytes	Matrix entries a, b, c, d (uint8)
37 .. 44	Original Size	8 Bytes	uint64 big-endian original unpadded file size
45 .. 46	Filename Length	2 Bytes	uint16 big-endian filename byte count (flen)
47 .. 47+flen	Filename	Variable	UTF-8 encoded original file name
+00 .. +07	Ciphertext Len	8 Bytes	uint64 big-endian DES ciphertext length (clen)
+08 .. +08+clen	DES Ciphertext	Variable	PKCS#7 padded encrypted payload bytes
-32 .. End	HMAC-SHA256 Tag	32 Bytes	Authentication signature over all preceding bytes

8. Implementation & Tech Stack Architecture

SecureVault is structured as a clean, decoupled monorepo designed to run locally on standard hardware without external cloud services:

- **Frontend Architecture:** Built with Next.js 16 (React 19, TypeScript), Tailwind CSS, and Framer Motion. Features an Apple-inspired Liquid Glass UI, responsive dark/light mode toggle with localStorage persistence, file dropzone with touch support, real-time pipeline visualizers, and an interactive live Hill matrix invertibility calculator.
- **Backend Architecture:** Built with Python 3 and FastAPI, served via Uvicorn. Implements path traversal defense, zero plaintext password logging, constant-time authentication, and modular crypto packages.
- **Persistence Layer:** Lightweight local SQLite database (database.db) recording audit history, file sizes, timestamps, and SHA-256 hashes without storing user passwords or keys.

9. Experimental Verification & Test Suite

The system has been verified using an automated test suite comprising **18 test cases** across both unit and integration levels. All 18 tests pass with 100% success:

Test Case Category	Target Data / Condition	Verification Standard	Result
01. Hill Cipher Unit Test	Random 8-bit byte sequences in Z_256	Matrix modular inversion $C \cdot K^{-1} == P$	PASS
02. DES-CBC Unit Test	Binary blocks with PKCS#7 padding	Block alignment & padding removal	PASS
03. Plaintext Document	ASCII / UTF-8 text file (.txt)	SHA-256(Original) == SHA-256(Decrypted)	PASS
04. Binary Document	Portable Document Format (.pdf)	Header & binary EOF byte-for-byte equality	PASS
05. Raster Image	PNG / JPEG bitmap graphic (.png)	Pixel buffers & metadata byte-for-byte equality	PASS
06. Compressed Archive	Multi-file ZIP archive (.zip)	Archive integrity & subfile extraction verified	PASS
07. Edge Case: Empty File	0-byte empty file	Graceful handling with 0-byte roundtrip	PASS
08. Large Binary Dataset	1 MB pseudo-random binary payload	Hash match across 1,048,576 bytes	PASS
09. Wrong Password Rejection	Incorrect credentials supplied	HMAC verification fails; 400 Bad Request	PASS
10. Ciphertext Tampering	Single-bit flipped in ciphertext	HMAC fails; DES unpadding aborted safely	PASS
11. Corrupted Container	Truncated .svault file	Container parser detects incomplete header	PASS
12. REST API Integration	Upload -> Encrypt -> Download -> Decrypt	Full roundtrip via HTTP client endpoints	PASS

10. Slide Deck Blueprint (Direct Directive for NotebookLM PPT Generation)

Note for NotebookLM: Use the following 18-slide structure as the primary blueprint when generating presentation outlines, slide content, and presentation summaries. Maintain large typography, high contrast, and minimal colors.

Slide 01: Title Slide: SECURE FILE SHARING SYSTEM Using Hill Cipher + DES (SecureVault). B.Tech CSE Mini-Project, CNS Course.

Slide 02: Why Secure File Sharing?: Problem statement: 01 Single-layer vulnerability, 02 Classical Hill Cipher cannot handle binary files (mod 26 limit), 03 DES has limited 56-bit key. Our Approach: Hill in Z₂₅₆ + DES-CBC + HMAC-SHA256.

Slide 03: Challenge vs. SecureVault: Comparative analysis: Left (Challenge: mod 26 restriction, single cipher risk) vs. Right (SecureVault: Z₂₅₆ byte handling, CBC mode, 100k PBKDF2 rounds, HMAC tag).

Slide 04: System Architecture Pipeline: Horizontal pipeline: [01 FILE] -> [02 PBKDF2] -> [03 HILL] -> [04 DES-CBC] -> [05 HMAC] -> [06 .SVAULT]. Fully reversible decryption.

Slide 05: Cryptographic Key Pipeline: User password + 16B salt -> PBKDF2 (100k rounds) -> 64 bytes partitioned into DES Key (8B), HMAC Key (32B), and Hill Matrix Seed (4B).

Slide 06: Hill Cipher in Ring Z₂₅₆: Central formula: $C = (P * K^A) \bmod 256$. 3 points: operates on 8-bit bytes, supports arbitrary files, invertible 2x2 matrix.

Slide 07: Hill Cipher Invertibility: Equation: $\gcd(\det(K), 256) = 1$. Since $256 = 2^8$, $\det(K)$ MUST BE ODD. Parity construction proof: diagonal odd, off-diagonal even guarantees odd determinant.

Slide 08: DES in CBC Mode: Specifications: 64-bit blocks + 56-bit key + 16 Feistel rounds. CBC chaining formula $C_i = \text{DES}(P_i \oplus C_{i-1})$ with random 8-byte IV.

Slide 09: Why CBC Over ECB?: Comparison: ECB mode leaks structural patterns (identical blocks = identical ciphertexts). CBC mode diffuses blocks using XOR chaining.

Slide 10: Defense in Depth (PBKDF2 + HMAC): Two security pillars: PBKDF2 (100k rounds mitigates GPU dictionary attacks) and HMAC-SHA256 (constant-time verification stops bit-flipping).

Slide 11: .svault Binary Container: Visual layout: Magic b'SVAULT', Version, Salt, IV, Hill Matrix, Metadata, DES Ciphertext, HMAC-SHA256 tag.

Slide 12: Important Security Advisory: Prominent statement: DES IS OBSOLETE FOR MODERN PRODUCTION SECURITY. 56-bit key is brute-force vulnerable. Used here for academic Feistel study. Modern standard: AES-256-GCM.

Slide 13: Testing & Quality Assurance: Display numbers: 18 TEST CASES, 100% ALL PASSED. Checkmarks: TXT, PDF, PNG/JPG, ZIP, Empty file, 1MB binary, Wrong password, Tampered ciphertext, Corrupted container.

Slide 14: Byte-for-Byte Validation: Original File SHA-256 == Decrypted File SHA-256 -> ✓ BYTE-FOR-BYTE MATCH. Proves complete mathematical reversibility.

Slide 15: Live Demonstration Flow: 7-step walkthrough: 1 Upload -> 2 Set Key -> 3 Hill mod 256 -> 4 DES-CBC -> 5 Get .svault -> 6 Decrypt -> 7 SHA-256 Match ✓.

Slide 16: Viva Defense: Core Questions: Cheat sheet: Why Z₂₅₆? (byte compatibility); Why odd determinant? (invertibility); Why CBC? (pattern elimination); Why HMAC? (integrity); Why not DES in production? (56-bit key).

Slide 17: SecureVault Key Takeaways: Summary: 01 Binary-compatible Hill, 02 Layered Feistel defense, 03 PBKDF2 + HMAC integrity, 04 Verified 100% byte fidelity.

Slide 18: Final Slide: THANK YOU. Questions & Answers. SecureVault • Department of Computer Science & Engineering.

11. Viva Voce Defense Question & Answer Guide

This section prepares students for oral examination and technical defense before faculty panels:

Q1: Can classical Hill Cipher encrypt images or PDF files without corruption?

Answer: No. Classical Hill Cipher operates modulo 26 on uppercase English letters (A-Z). Binary files contain 256 distinct byte values (0-255). SecureVault formulates the Hill Cipher over Ring Z_{256} , mapping every byte directly to an element of the ring, preserving binary headers and data byte-for-byte.

Q2: How do you mathematically guarantee that the Hill matrix is invertible modulo 256?

Answer: In any integer ring Z_m , an $n \times n$ matrix is invertible if and only if $\gcd(\det(K) \bmod m, m) = 1$. Since $256 = 2^8$, its only prime factor is 2. Therefore, $\gcd(\det(K), 256) = 1$ if and only if $\det(K)$ is an ODD integer. SecureVault enforces that diagonal entries a and d are odd, and off-diagonal entries b and c are even. Since $(\text{odd} \times \text{odd}) - (\text{even} \times \text{even}) = \text{odd} - \text{even} = \text{odd}$, $\det(K)$ is mathematically guaranteed to be odd, ensuring universal invertibility.

Q3: What attack does Encrypt-then-MAC (HMAC-SHA256) prevent?

Answer: Symmetric ciphers in CBC mode provide confidentiality, but not authenticity or integrity. An attacker could alter ciphertext bits (bit-flipping) or conduct padding oracle attacks by observing server padding error responses. SecureVault computes an HMAC-SHA256 tag over the entire container and verifies it in constant time (`hmac.compare_digest`) prior to executing any decryption or unpadding. If corrupted, decryption aborts immediately with a generic error.

Q4: Why is CBC mode preferred over ECB mode in DES?

Answer: In ECB mode, each 64-bit block is encrypted independently. Identical plaintext blocks produce identical ciphertext blocks, preserving structural patterns (e.g. the ECB Penguin). In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption, initialized with a random 8-byte IV. Two identical files or blocks produce completely randomized, dissimilar ciphertexts.

Q5: Why is PBKDF2 used instead of a standard single-round SHA-256 hash?

Answer: Single-round SHA-256 hashes can be calculated billions of times per second on modern consumer GPUs, making dictionary attacks easy. PBKDF2 (NIST SP 800-132) applies 100,000 iterations of HMAC-SHA256 with a 16-byte random salt, making brute-force password guessing computationally intractable and preventing precomputed rainbow table attacks.

Q6: Is DES considered secure for modern production systems?

Answer: No. DES has an effective key length of 56 bits ($2^{56} \approx 7.2 \times 10^{16}$ keys), which can be exhausted by modern GPU clusters or distributed cloud hardware in a matter of hours. We explicitly disclose this in our application and project report. DES is included here strictly for academic exploration of Feistel structures and CBC block chaining. Modern production systems must deploy AES-256-GCM or ChaCha20-Poly1305.

12. Conclusion & Summary

SecureVault demonstrates that classical linear matrix algebra (Hill Cipher) and standard Feistel block ciphers (DES) can be rigorously compounded into a functional, authenticated file sharing system. By extending the Hill Cipher to the integer ring $\mathbb{Z}_{256}$, the system achieves universal binary compatibility with guaranteed mathematical invertibility. The end-to-end integration of PBKDF2, DES-CBC, PKCS#7 padding, and HMAC-SHA256 ensures complete byte-for-byte fidelity while providing defense-in-depth against structural pattern leakage, dictionary attacks, and unauthorized tampering.